

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			1. Oxygen crosses the membrane by (simple) diffusion (1) NOT facilitated 2. oxygen passes through the phospholipid (bilayer) (1) Membrane = neutral 3. Sodium ions pass through {channel/carrier} proteins (1) intrinsic proteins = neutral 4. by {facilitated diffusion/active transport/ cotransport} (1) Reject active transport in relation to channel proteins If oxygen and sodium are not treated separately, allow one mark max for correct linkage of diffusion with phospholipid bilayer or {facilitated diffusion/ active transport/ cotransport} with {channel/ carrier} proteins	4			4		
	(b)			1. The percentage ion composition of {fresh water/ habitat} is <u>lower</u> than that in the carp's blood plasma/ fresh water is hypotonic to the blood/ ORA (1) 2. The water potential of fresh water is {higher/less negative} than the water potential of the carp's blood/ ORA (1) 3. The carp takes in water (through the gills) by osmosis (1) 4. (Producing large volumes of dilute urine) prevents {osmotic lysis/ bursting} of cells (1) 5. Maintaining ion composition (1) Accept reduces the loss of ions		1 1	1 1	4		

Question				Marking details	Marks available					
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5	(c)	(i)		1. Percentage ion composition of sea water is <u>greater</u> than that of the blood plasma/ sea water is hypertonic/ water potential in the fish is <u>higher</u> than the sea water/ ORA (1) 2. ions will move into the blood plasma, {by (facilitated) diffusion/ down a concentration gradient/ or description of}(1) 3. therefore ions must be actively transported out (of the blood plasma into the surrounding sea water) (1)		1 1	1	3		
	(c)	(ii)		Cell 1 is from the salmon in sea water (1) Any one from: as it is {highly folded to give a larger surface} / contains {many/more} mitochondria to {provide ATP/ for respiration/ for active transport } (1)			1 1	2		

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5	(d)			1. As the percentage ion composition of the blood plasma is higher than the surrounding water (1) 2. Ions will move out of the {fish/gills/blood plasma} by {(facilitated) diffusion/down a concentration gradient} (1) 3. Acidification will {denature/ inactivate/ change the shape of} the (carrier/ channel) proteins in the specialised cells/ denature/ inactivate enzymes (involved in ATP production) (1) 4. Therefore {no/fewer} ions are (actively) transported <u>into</u> the {fish/gills/blood (plasma)}/ fewer ions can diffuse out (1) (in context of damaged proteins)		1	1	4		
				Question 5 total	4	5	8	17	0	0